

Static liquefaction of sand in plane strain

Dariusz Wanatowski and Jian Chu

Abstract: Experimental results on the static liquefaction behaviour of sand under plane-strain conditions are presented in this paper. Undrained tests on very loose sand under both plane-strain and axisymmetric conditions were conducted and the results compared. The test data show that the undrained behaviour of sand under plane strain is similar to that under axisymmetric conditions. However, the critical-state line (CSL) on both the q - p' and the e - p' planes determined under plane-strain conditions is different from that under axisymmetric conditions. The slope of the CSL is different as a result of the influence of the intermediate principal stress. The state parameter (ψ), which is measured with reference to the CSL in the e - $\log p'$ plane, is also different: the ψ value for plane strain is about 0.05 less than that for axisymmetric conditions for the sand tested. The instability behaviour of very loose sand under undrained plane-strain conditions is also studied. Based on the testing data, a unique relationship between the stress ratio of the instability line and ψ is established to enable the triaxial results to be used for plane-strain conditions.

Key words: failure, sands, laboratory tests, liquefaction, shear strength.

Résumé : Dans cet article, on présente des résultats expérimentaux pour étudier le comportement en liquéfaction statique du sable dans des conditions de déformation plane. On a réalisé et comparé des essais non drainés sur du sable très meuble dans des conditions de déformation plane et axisymétriques. Les données des essais montrent que le comportement non drainé du sable dans des conditions de déformation plane est similaire à celui dans des conditions axisymétriques. Cependant, la ligne d'état critique « CSL » tant sur les plans q - p' que e - p' déterminée dans des conditions de déformation plane est différente de celle en conditions axisymétriques. La pente CSL, est différente en raison de l'influence de la contrainte principale intermédiaire. Le paramètre d'état, ψ , qui est mesuré en référence à la CSL dans le plan e - $\log p'$, est également différent. La valeur de ψ pour la déformation plane est d'environ 0,05 plus faible que pour les conditions axisymétriques pour le sable testé. On étudie également le comportement de l'instabilité du sable très meuble dans des conditions de déformation plane non drainée. Sur la base des données des essais, on établit une relation unique entre le rapport de contraintes de la ligne d'instabilité et le paramètre d'état pour permettre d'utiliser les résultats des essais triaxiaux pour les conditions de déformation plane.

Mots-clés : rupture, sables, essais en laboratoire, liquéfaction, résistance en cisaillement.

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Introduction

The term “static liquefaction” usually refers to a phenomenon in which loose sand loses a large percentage of its shear resistance and flows in a liquid manner when it is subjected to undrained monotonic loading (Castro 1969). The static liquefaction of sand is often considered a trigger for the failure of loose granular slopes (Casagrande 1965). Such flow failures have occurred at the Calaveras Dam (Hazen 1918); the Fort Peck Dam (Casagrande 1965); Helsinki Harbour (Andresen and Bjerrum 1976); the Nerlerk berm (Sladen et al. 1985b); and the Wachusett Dam (Olson and Stark 2003). Similar failures occurred at the Aberfan coal waste tip in Wales (HMSO 1967), a gypsum tailings impoundment

in eastern Texas (Kleiner 1976), and the Merriespruit tailings dam in South Africa (Fourie et al. 2001). Flow failures are characterized by a sudden loss of strength followed by a rapid development of large deformations, so their effects are often dramatic and devastating. As a result, numerous studies have been carried out to understand the liquefaction phenomenon of granular soils (e.g., Castro 1969; Green and Ferguson 1971; Poulos 1981; Sladen et al. 1985a; Ishihara 1993; Lade 1993; Yamamuro and Lade 1997; Leong et al. 2000; Yang 2002; Been and Jefferies 1985, 2004). These studies demonstrated that when loose sand is subjected to undrained loading, the deviator stress will peak at a small strain and then decrease substantially in the post-peak region until an ultimate state is reached at large strains. The ultimate state is also referred to as the steady state (Poulos 1981). Studies have shown that the ultimate state is practically the same as the critical state and that it is unique for a given soil (Poulos 1981; Alarcon-Guzman et al. 1988; Hird and Hassona 1990; Been et al. 1991; Negussey and Islam 1994; Chu 1995).

Most of the above-cited studies were carried out under axisymmetric conditions. However, most failures can only be simplified into plane-strain conditions. Although several studies have been carried out on the behaviour of sand under plane-strain conditions (e.g., Han and Vardoulakis 1991;

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Finno et al. 1996, 1997; Mokni and Desrues 1999; Finno and Rechenmacher 2003; Desrues and Viggiani 2004), data on liquefaction under plane-strain conditions are still very limited. Further studies are required to establish the differences in liquefaction behaviour under axisymmetric conditions and under plane-strain conditions and also to determine whether the findings on liquefaction established under axisymmetric conditions are directly applicable to plane-strain problems.

The objectives of this study were to observe the static liquefaction behaviour of loose sand under plane-strain conditions and to compare the undrained behaviour of loose sand under axisymmetric conditions with that under plane-strain conditions. A new plane-strain apparatus was adopted in this study and used to conduct K_0 consolidated undrained (CK_0U) tests on sand under both plane-strain and axisymmetric conditions. These experimental results will help to explain the liquefaction behaviour of loose sand under plane-strain conditions and the limitations of the liquefaction studies under axisymmetric conditions. Researchers have introduced several important concepts and parameters to describe the stress-strain behaviour of soil. These include the critical-state or steady-state concept and the critical-state line (CSL) (Schofield and Wroth 1968; Castro 1969; Poulos 1981); the instability line, which was defined by Lade (1992) to describe the conditions under which a soil may become unstable; the b value ($b = (\sigma_2 - \sigma_3)/(\sigma_1 - \sigma_3)$), which is used to describe the influence of intermediate principal stress; and the state parameter ψ , which was proposed by Been and Jefferies (1985) to measure the initial stress and void ratio (e) of soil. The data obtained in this study also will be used to examine some fundamental aspects of the above concepts and parameters: (1) how the b value or the three principal stresses vary during a plane-strain test and whether a constant b value can be assumed for plane-strain conditions; (2) whether the CSL is affected by the b value, and if it is, how the critical-state parameters vary from axisymmetric to plane-strain conditions or are affected by the intermediate principal stress or b value; (3) whether the state parameter, ψ , can be uniquely defined, and if not, whether it is also dependent on the intermediate principal stress; and (4) whether an instability line can also be defined under plane-strain conditions and how it is affected by the b value.

Experimental arrangement

The plane-strain apparatus is shown in Fig. 1, and the complete testing system is presented in Fig. 2. The prismatic specimen is 120 mm high and 60 mm $\times$ 60 mm in cross section. The plane-strain condition was imposed by using two pairs of horizontal tie rods to fix two stainless steel vertical platens, 35 mm thick $\times$ 74 mm wide $\times$ 120 mm high, in position (see Fig. 1a). Two submersible pressure transducers on each vertical platen (see Fig. 1d) were used to measure the lateral stress (σ_2). These elevated-diaphragm soil-pressure transducers, manufactured by Kyowa Electronic Instruments Co., Ltd., measured the lateral pressures at both the top and the bottom positions on each side of the specimen. The four lateral pressure measurements in almost all the tests were essentially the same before the occurrence of

shear bands. Therefore, the lateral pressure was evaluated as the average of the four readings from the four transducers. The outer diameter of the transducers was 30 mm, and the diameter of the sensing diaphragms was 23 mm. The transducers had a capacity of 1000 kPa and a resolution of 0.01 kPa. The pressure transducers were calibrated by placing the vertical platens into a triaxial cell and compressing them using hydrostatic pressure. The maximum error determined from the linear regression during the calibration was 0.5 kPa (i.e., 0.05% of the maximum capacity). The vertical load was applied by a force actuator (GDS Instruments Ltd., Hook, Hampshire, UK) mounted at the bottom of a loading frame. A computer controlled the movement of the base pedestal to achieve the desired load rate or rate of displacement. The vertical load was measured with a 10 kN submersible load cell. The cell pressure (σ_3) was applied through a digital pressure-volume controller (DPVC). Another DPVC was used either to control the back pressure from the bottom of the specimen while measuring the volumetric change in a drained test or to control the volumetric change while measuring the pore-water pressure (u) in an undrained test. The resolutions of the DPVCs were 0.5 kPa for pressure and 0.5 mm³ for volume change. A pressure transducer with a capacity of 1000 kPa and a resolution of 0.01 kPa was also used to record u at the top of the specimen.

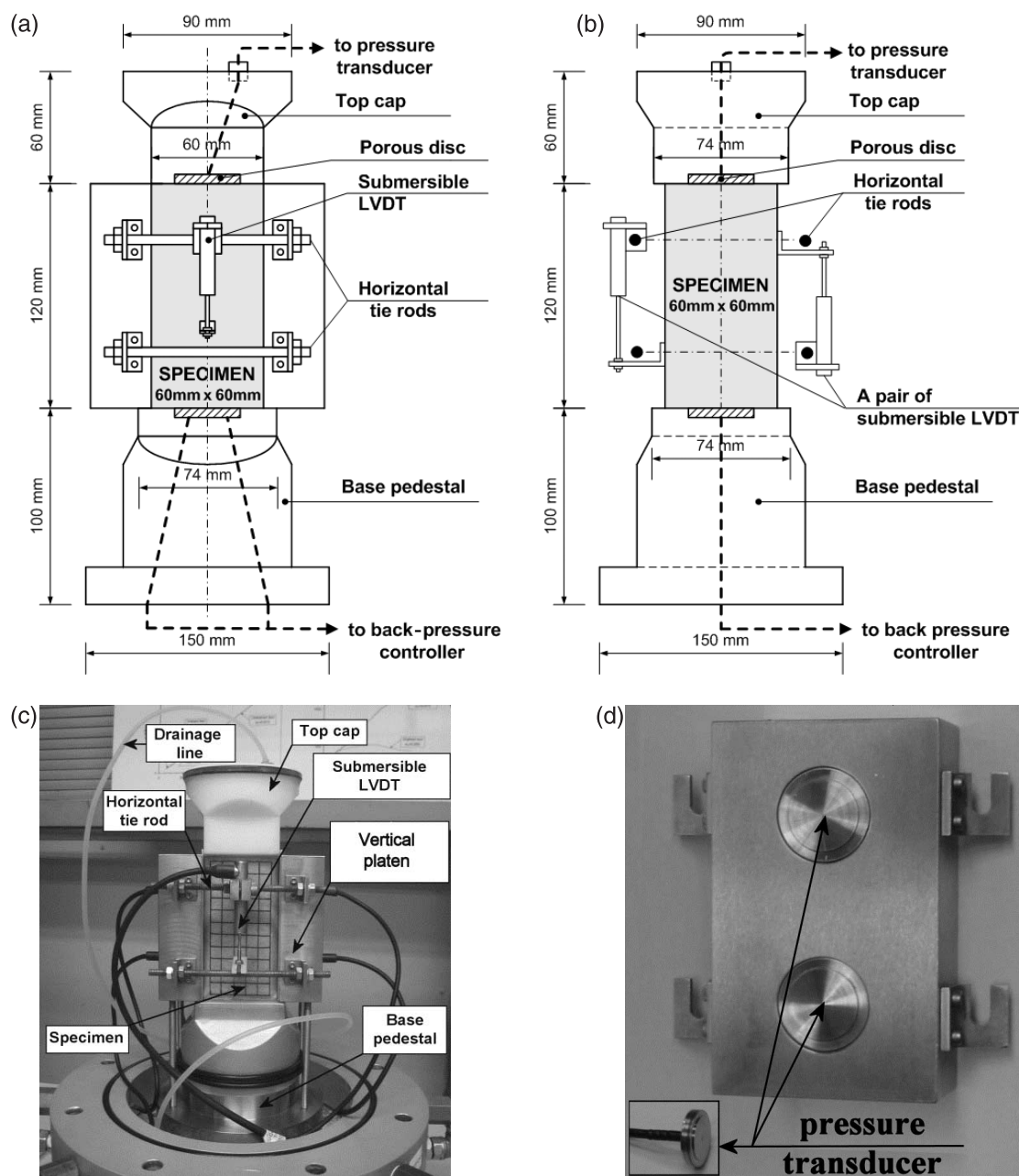
The free-end technique (Rowe and Barden 1964) was used to properly enlarge and lubricate all rigid platens to reduce the boundary frictions and to delay the occurrence of nonhomogeneous deformations. The platens were enlarged by a factor of 1.23 times the specimen dimension and lubricated with a high-vacuum silicone grease. The free end for the top cap and the base pedestal consisted of a layer of high-vacuum silicone grease and a sheet of latex, which was 1.3–1.5 times thicker than the mean grain size. The free end for the vertical loading platen had a layer of grease and a 1 mm thick Teflon sheet. These free-end arrangements have been verified to be effective for both triaxial and multiaxial tests (Chu and Lo 1993; Chu et al. 1996).

The vertical displacement was measured with a pair of miniature submersible linear variable differential transformers (LVDTs). The internal LVDT had a maximum range of ± 5 mm and a resolution of 0.1 μ m. The maximum error obtained within the linear regression range was 0.002 mm (or 0.0017% axial strain, ϵ_1). The readings from the external LVDT were used when the internal LVDT went out of the linear range at $\epsilon_1 \approx 6\%$. The external LVDT, mounted to the vertical loading shaft (Fig. 2), could travel 50 mm and had a resolution of 0.01 mm.

As shown in Fig. 1, the top cap and bottom pedestal of the plane-strain apparatus were designed to permit a transition from a square to a circular cross section, so a cylindrical membrane could be used. The membrane had a thickness of 0.4 mm and a circumference of 240 mm. The liquid rubber technique (Lo et al. 1989) was adopted to reduce the effect of membrane penetration, that is, the inner side of the membrane was coated with a thin layer of liquid rubber before the placement of sand. Plane-strain tests commenced only after the liquid rubber had hardened.

As mentioned earlier, σ_2 was measured with four submersible pressure transducers. The onset of nonhomoge-

Fig. 1. Plane-strain apparatus: (a) elevation view; (b) side view (without vertical platens); (c) photograph of elevation; (d) photograph of vertical platen. LVDT, linear variable differential transformer.



neous deformation was identified as the point where the four σ_2 values started to diverge. Although several plane-strain devices that measure σ_2 have been developed (e.g., Cornforth 1964; Tatsuoka et al. 1986; Topolnicki et al. 1990; Finno et al. 1996), the use of σ_2 measurements for the detection of shear bands has not been reported before.

The triaxial experiments were carried out using a fully automated triaxial testing system. The specimen was 200 mm in height and 100 mm in diameter. The 10 kN submersible load cell was again used to measure the vertical load. The confining pressure was controlled with a DPVC, and another DPVC controlled the back pressure from the bottom of the specimen while simultaneously measuring the volume change. Pressure transducers measured u at the top and the

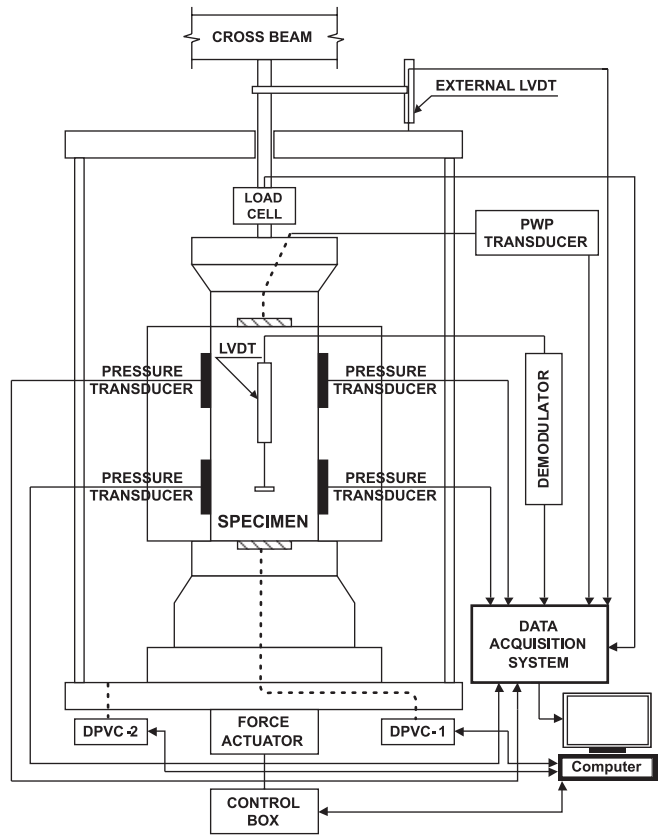
base of the specimen. For details of the triaxial testing arrangement, see Leong et al. (2000).

Tested soil and specimen preparation

Basic properties of tested sand

Changi sand, used for the Changi land reclamation project in Singapore, was tested in this study. This marine-dredged silica sand has also been used in previous studies on liquefaction and instability under axisymmetric conditions (Leong et al. 2000; Chu and Leong 2002; Leong and Chu 2002; Chu et al. 2003). The sand grains were mainly sub-angular, and the shell content was between 12% and 14%. The grain-size distribution curve is shown in Fig. 3. The ba-

Fig. 2. Plane-strain testing system. DPVC, digital pressure–volume controller; LVDT, linear variable differential transformer; PWP, pore-water pressure.



sis physical properties of the Changi sand are summarized in Table 1.

Specimen preparation

The moist tamping method was used to prepare all specimens. The sand was premixed to a moisture content of 5% and then deposited into a mould in six layers. Each layer was compacted with a small tamper, and the number of blows applied to each layer was carefully controlled. The compactive effort was increased toward the top of the specimen to achieve greater uniformity. For plane-strain specimens, the two vertical platens were assembled by tightening the horizontal tie bars so that the readings of the pressure transducers in the platens were just about to increase.

Test procedures and test program

A cell pressure of 20 kPa was applied to the specimen after it was placed in the apparatus. The specimen was then flushed with de-aired water from the bottom to the top for approximately 60 min under a water head of 0.5 m. After that, both the cell pressure and the back pressure were increased simultaneously while the effective stress was held constant at 20 kPa, until the back pressure reached 350 kPa. The Skempton’s *B* value was then checked. A *B* value of 0.96–0.98 was achieved for all specimens. The cell pressure and the back pressure were then increased to 420 and 400 kPa, respectively. The void ratio (*e*) after saturation was

Fig. 3. Grain-size distribution curve for Changi sand.

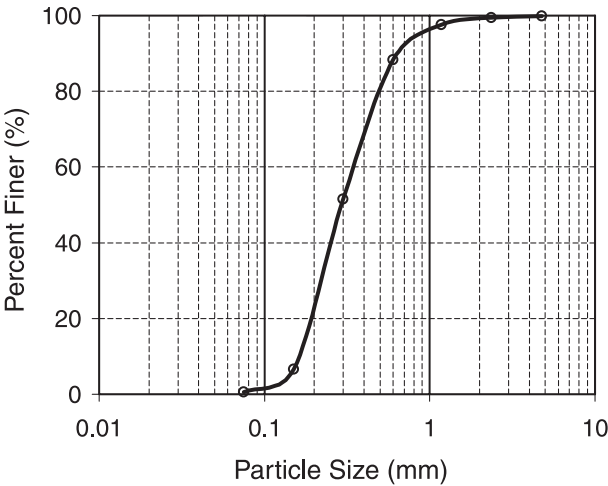


Table 1. Basic properties of tested sand.

Mean grain size (mm)	0.30–0.35
Uniformity coefficient	2
Specific gravity	2.6
Maximum void ratio ^a	0.916
Minimum void ratio ^b	0.533
Fines content (%)	0.4
Shell content (%)	12–14

^aDetermined according to ASTM D4254.

^bDetermined according to ASTM D4253.

back-calculated using the water content of the specimen at the end of test and the volume changes during consolidation and shearing, a procedure proposed by Verdugo and Ishihara (1996).

All specimens were consolidated from an initial isotropic stress state of 20 kPa to the required stress state along the *K*₀ path. The *K*₀ consolidation in both the triaxial and the plane-strain tests was carried out by maintaining the *dε_v/dε₁* ratio at one, a strain-path method described by Lo and Chu (1991). The *K*₀ consolidation for each specimen was carried out under a deformation-controlled loading mode at a constant rate of 0.05 mm/min. At the end of *K*₀ consolidation, the stresses on the specimen were kept constant for an aging period of 30 min. A summary of the plane-strain and triaxial tests is given in Table 2. The estimated value for the state parameter (*ψ*, as proposed by Been and Jefferies 1985) for each test is also given in Table 2. Although each specimen can be classified as very loose according to Skempton (1986), the *ψ* values for some tests were negative.

Experimental results

Drained behaviour in triaxial tests

The typical *q*–*ε₁* and *ε_v*–*ε₁* curves for isotropically consolidated drained (CID) tests on very loose sand are shown in Figs. 4*a* and 4*b*, respectively. The deviatoric stress (*q*) and the volumetric strain (*ε_v*) approached a constant value at the end of each test. Therefore, the failure line obtained for very loose sand as shown in Fig. 4*c* is also the CSL for triaxial conditions (CSL_{tc}). The slope of the CSL_{tc} (*M_{tc}*) on the *q*–*p*′ plane

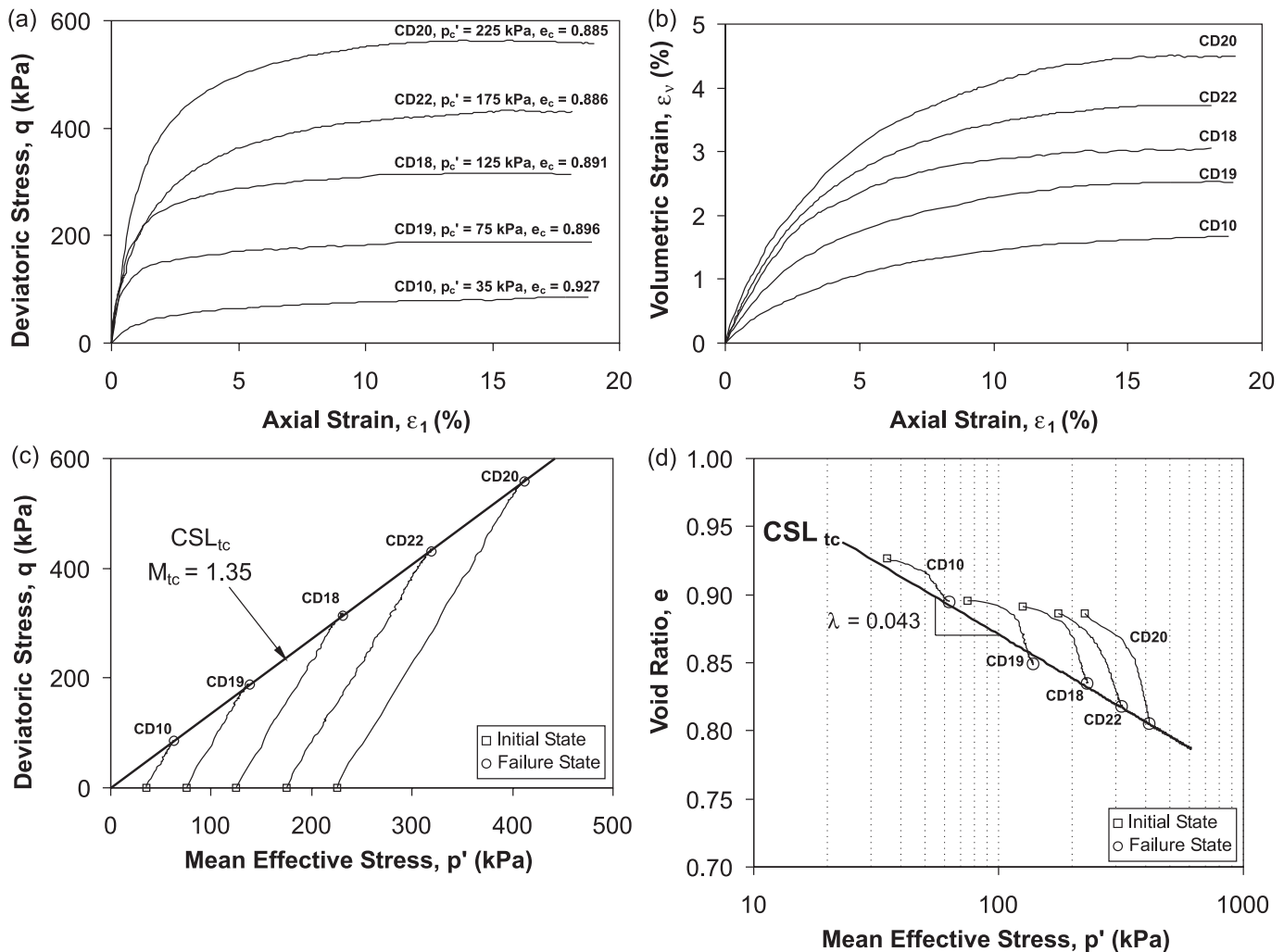
Table 2. List of tests conducted.

Test No.	Type of test	Consolidated state				Peak state ^a				End of test			
		e_c	p'_c (kPa)	q_c (kPa)	e_p	p'_p (kPa)	q_p (kPa)	b_p	e_f	p'_f (kPa)	q_f (kPa)	b_f	ψ
CD10	CID-tc	0.927	35	0	0.895	63	86		0.895	63	86		0.009
CD19	CID-tc	0.896	75	0	0.849	138	188		0.849	138	188		0.012
CD18	CID-tc	0.891	125	0	0.835	231	314		0.835	231	314		0.031
CD22	CID-tc	0.886	175	0	0.818	319	431		0.818	319	431		0.041
CD20	CID-tc	0.885	225	0	0.805	412	558		0.805	412	558		0.053
CU44	CIU-tc	0.972	150	0	0.972	101	73		0.972	8	6		0.108
CU20	CIU-tc	0.916	150	0	0.916	107	95	0.00	0.916	4	1	0.00	0.052
CU26	CIU-tc	0.888	150	0	0.888	116	123		0.888	3	5		0.024
CU21	CIU-tc	0.864	150	0	0.864	124	168		0.864	124	168		0.000
CU27	CIU-tc	0.844	150	0	0.844	202	278		0.844	202	277		-0.020
K0U3	CK ₀ U-tc	0.922	199	146	0.922	197	169		0.922	15	21		0.058
K0U1	CK ₀ U-tc	0.899	191	151	0.899	188	186		0.899	66	91		0.035
K0U4	CK ₀ U-tc	0.881	200	182	0.881	188	211		0.881	106	144		0.017
K0U2	CK ₀ U-tc	0.844	201	202	0.844	197	244		0.844	189	255		-0.019
K0D11ps	CK ₀ D-ps	0.963	35	25	0.946	50	57	0.30	0.946	50	57	0.30	0.005
K0D07ps	CK ₀ D-ps	0.932	99	70	0.894	162	184	0.29	0.894	162	184	0.29	0.017
K0D08ps	CK ₀ D-ps	0.914	202	166	0.869	306	358	0.28	0.869	306	358	0.28	0.027
K0D09ps	CK ₀ D-ps	0.900	298	275	0.854	434	505	0.28	0.854	434	505	0.28	0.030
K0U04ps	CK ₀ U-ps	0.935	97	70	0.935	91	80	0.40	0.935	11	14	0.40	0.018
K0U05ps	CK ₀ U-ps	0.915	198	157	0.915	189	166	0.21	0.915	70	75	0.37	0.031
K0U06ps	CK ₀ U-ps	0.899	298	245	0.899	291	259	0.26	0.899	165	189	0.35	0.029
K0U08ps	CK ₀ U-ps	0.902	198	168	0.902	196	190	0.28	0.902	94	108	0.28	0.018
K0U09ps	CK ₀ U-ps	0.893	198	174	0.893	199	202	0.29	0.893	137	161	0.29	0.009
K0U10ps	CK ₀ U-ps	0.888	199	186	0.888	195	215	0.31	0.888	152	177	0.31	0.004
K0U11ps	CK ₀ U-ps	0.884	199	191	0.884	189	221	0.34	0.884	186	218	0.34	0.000
K0U12ps	CK ₀ U-ps	0.868	198	205	0.868	259	318	0.29	0.868	232	258	0.27	-0.016

Note: ps, plain strain; tc, triaxial compression.

^aFailure state in drained tests.

Fig. 4. Drained behaviour of very loose sand in triaxial compression: (a) stress–strain curves; (b) ε_v – ε_1 curves; (c) effective stress paths; (d) state plot and CSL_{tc} on $e - \log p'$ plane. CSL_{tc} , critical-state line under triaxial compression; e_c , void ratio at end of consolidation; M_{tc} , effective stress ratio at critical state for triaxial compression; p'_c , mean effective stress at end of consolidation; λ , slope of CSL on $e - \log p'$ plane.



is 1.35, which corresponds to a friction angle $((\phi_{cs})_{tc})$ of 33.4° . The CSL_{tc} on the $e - \log p'$ plane is plotted in Fig. 4d.

Undrained behaviour in triaxial tests

The results of five isotropically consolidated undrained (CIU) tests conducted on very loose specimens are shown in Fig. 5. Strain-softening behaviour was observed in tests CU44, CU26, and CU20, whereas strain-hardening behaviour was observed in test CU27. In test CU21, the deviator stress (q) and excess pore-water pressure (Δu) remained constant after the peak was attained, as shown in Figs. 5a and 5b, respectively. The void ratio at the end of consolidation (e_c) in test CU21 is 0.864 and is therefore the void ratio at critical state (e_{cs}) (Casagrande 1936). The value for e_{cs} sets the boundary between strain-hardening and strain-softening behaviour under the given mean effective stress (p'). Strain-softening behaviour was obtained for all tests with $e_c > e_{cs}$ (i.e., >0.864), and strain-hardening behaviour was observed for tests with $e_c < e_{cs}$ (Fig. 5a).

The effective stress paths for all the CIU tests are plotted in Fig. 5c. The CSL_{tc} as, obtained from the CID triaxial tests

(see Fig. 4c), is also plotted in Fig. 5c. It can be seen that the effective stress path varies with e_c . In addition, η_{IL} , which is the effective stress ratio (q/p') at peak deviatoric stress, was determined for the tests that exhibited strain softening (Fig. 5c) by drawing a line through the peak point of the undrained effective stress path. Lade (1992) called such a line the instability line. It can be seen from Fig. 5c that the smaller the e_c is, the higher the mobilized stress ratio at the peak (η_{IL}) will be, as discussed by Chu et al. (2003).

The results of four CK_0U triaxial tests conducted on very loose sand are presented in Fig. 6. The behaviour observed was similar to that obtained from the CIU tests, except the peak occurred at a much smaller axial strain (ε_1) than in the CIU tests. The effective stress paths are shown in Fig. 6c. It should be noted that when a true K_0 consolidation was imposed by using the strain path method, the K_0 value at the end of K_0 consolidation was affected by the void ratio of the soil, as discussed by Chu and Gan (2004). This may explain why the different CK_0U tests started from different stress states. Figure 6c also shows that mobilized η_{IL} increases with decreasing e_c .

Fig. 5. Results of consolidated isotropically undrained triaxial tests conducted on very loose sand: (a) stress–strain curves; (b) Δu – ε_1 curves; (c) effective stress paths. CID, consolidated isotropically drained tests; CSL_{tc} , critical-state line under triaxial compression; e_c , void ratio at end of consolidation; pwp, pore-water pressure; η_{IL} , effective stress ratio at peak deviatoric stress.

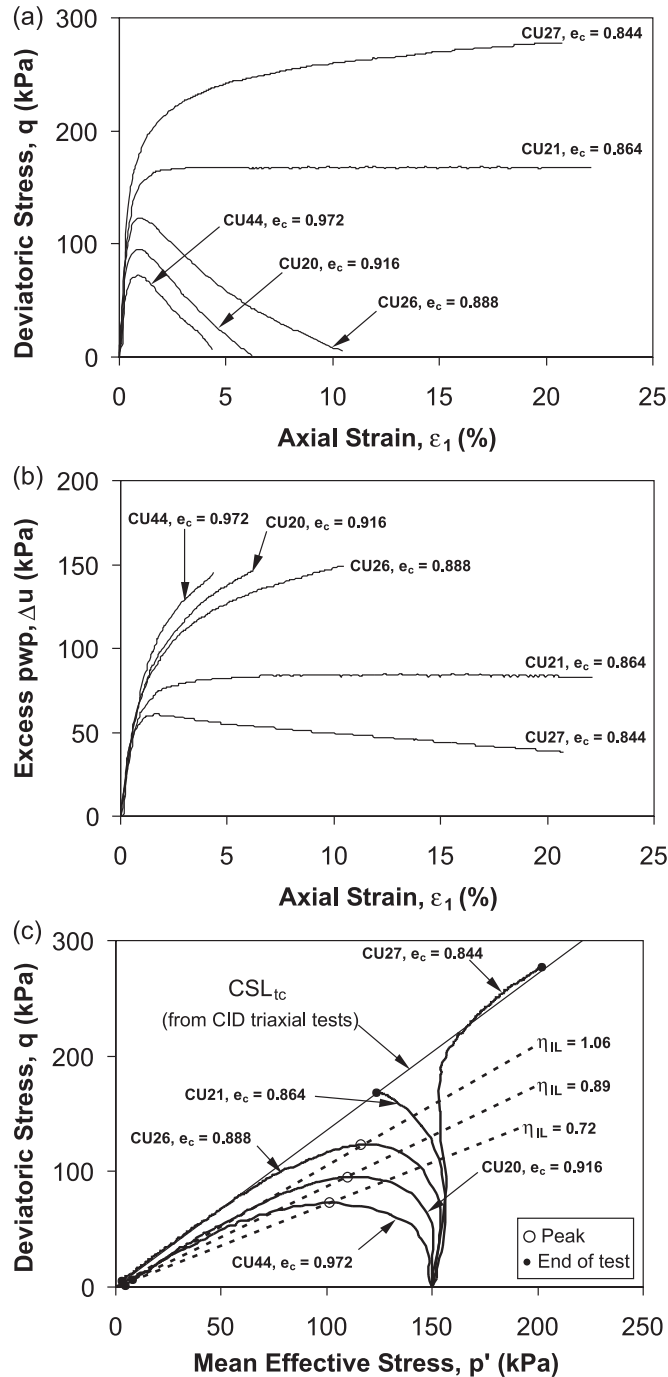
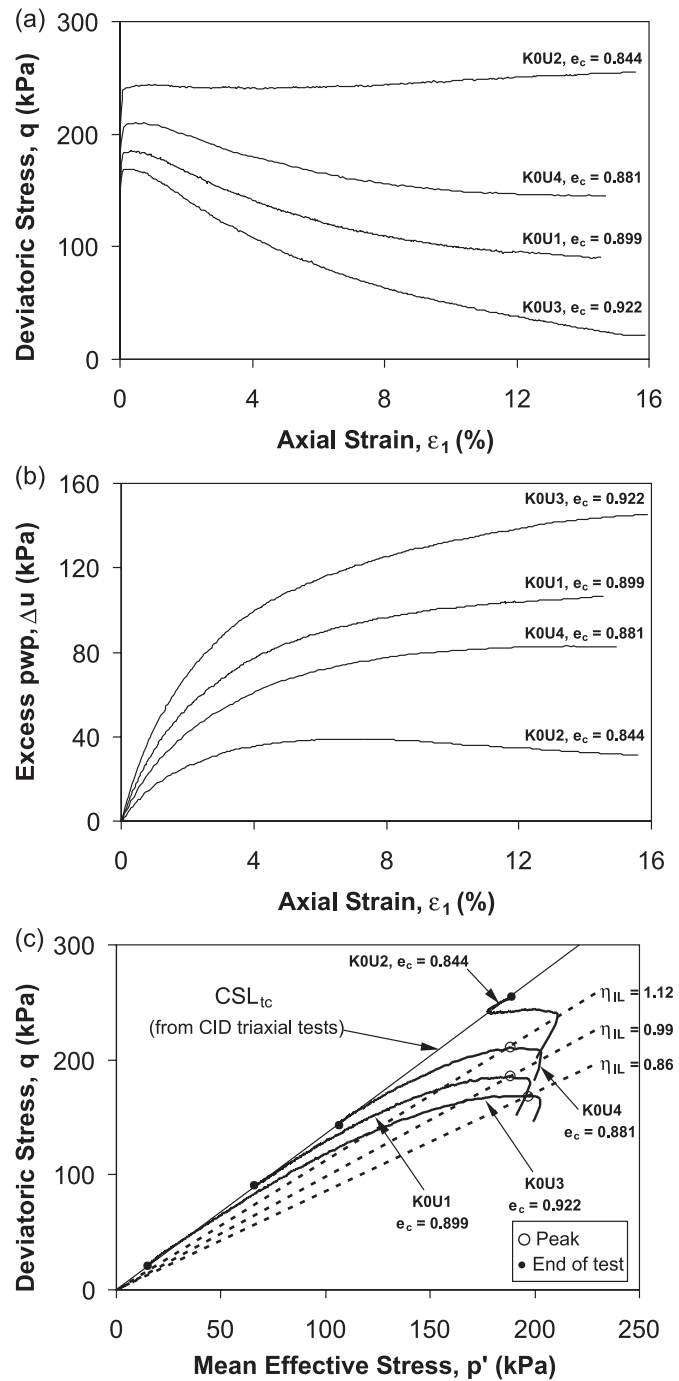


Fig. 6. Results of consolidated undrained triaxial tests conducted on very loose sand under conditions of no lateral strains: (a) stress–strain curves; (b) Δu – ε_1 curves; (c) effective stress paths. CID, consolidated isotropically drained tests; CSL_{tc} , critical-state line under triaxial compression; e_c , void ratio at end of consolidation; pwp, pore-water pressure; η_{IL} , effective stress ratio at peak deviatoric stress.

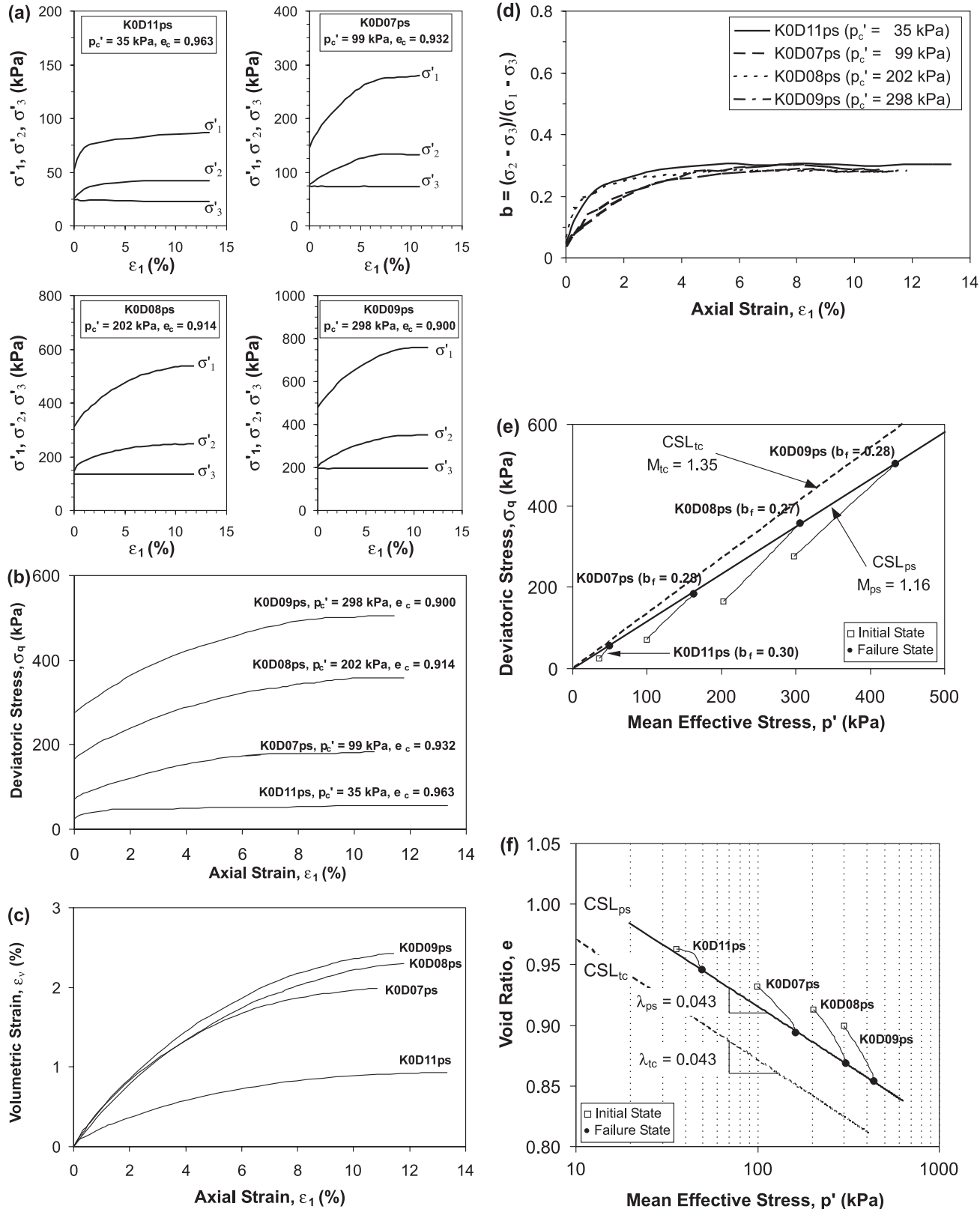


Drained behaviour in plane-strain tests

The results of four K_0 consolidated drained (CK₀D) plane-strain tests conducted on very loose sand specimens are presented in Fig. 7. The specimens were consolidated to mean effective stresses (p'_c) of 35, 99, 202 and 298 kPa; the void ratios after consolidation (e_c) were 0.939, 0.932, 0.914 and

0.900, respectively. The variation in the effective principal stresses is shown in Fig. 7a. The effective stresses, the deviatoric stress, and the volumetric strain approached constant values at the end of each test (Figs. 7a, 7b, and 7c, respectively). Therefore, the failure line obtained for very loose sand in Fig. 7e is also the CSL for plane-strain

Fig. 7. Drained behaviour of very loose sand in plane-strain compression: (a) effective principal stresses; (b) stress–strain curves; (c) ε_v – ε_1 curves; (d) b – ε_1 curves; (e) effective stress paths; (f) state plot and critical-state line on e – $\log p'$ plane. b_f , b value at failure; CSL_{ps} , critical-state line under plane-strain conditions; CSL_{tc} , critical-state line under triaxial compression; e_c , void ratio at end of consolidation; M_{ps} , effective stress ratio at critical state for plane-strain conditions; M_{tc} , effective stress ratio at critical state for triaxial compression; p'_c , mean effective stress at end of consolidation; λ_{ps} , slope of CSL_{ps} under plane-strain conditions; λ_{tc} , slope of CSL_{tc} under triaxial compression.



conditions (CSL_{ps}). The slope of the CSL_{ps} (M_{ps}) on the q - p' plane is 1.16, which corresponds to a critical-state friction angle ($(\phi_{cs})_{ps}$) of 36.0° . The CSL_{ps} on the e - $\log p'$ plane is plotted in Fig. 7f. For comparison, the CSL_{tc} obtained from drained triaxial tests (Figs. 4c and 4d) is also plotted in Figs. 7e and 7f. It can be seen that the CSL_{ps} differs from the CSL_{tc} . The curves of b value ($b = (\sigma_2 - \sigma_3)/(\sigma_1 - \sigma_3)$) versus axial strain (ϵ_1) are also plotted in Fig. 7d. The b value for each test gradually increased to a peak and then remained constant within a narrow range of 0.27–0.30. More discussion of the drained behaviour of Changi sand in plane-strain tests at other densities is presented in Wanatowski and Chu (2006).

Undrained behaviour in plane-strain tests

The results of six CK_0 U plane-strain tests (K0U05ps, K0U08ps, K0U09ps, K0U10ps, K0U11ps, and K0U12ps) are presented in Fig. 8. All specimens were consolidated to a p'_c of 198–199 kPa. The void ratios (e_c) after K_0 consolidation were in the range of 0.868–0.915. The variation in the effective principal stresses for each specimen is shown in Fig. 8a. Figure 8b shows that strain softening occurred when $e_c > 0.884$. In these tests, the peak (or strain softening) occurred at a very small ϵ_1 , as shown in Figs. 8a and 8b. Strain hardening occurred for tests in which $e_c < 0.884$. The deviatoric stress (σ_q) and the excess pore-water pressure (u) remained constant after the peak of 0.884 was reached, so the critical state void ratio (e_{cs}) was likely 0.884 for the p'_c used in the tests. These behaviours were consistent with those in the triaxial tests, except the e_{cs} was different (see Fig. 5).

Strain softening in K0U12ps occurred after strain hardening. However, it was different from the strain softening that occurred in K0U05ps, K0U08ps, K0U09ps, and K0U10ps. The peak in test K0U12ps was reached at an ϵ_1 of approximately 2%, whereas the peak points in the other tests were reached at much lower values, ranging from 0.2% to 0.4%. In addition, the peak in K0U12ps occurred in conjunction with the formation of shear bands. Visible shear bands were not observed in the other tests. Therefore, the strain softening observed in K0U12ps was due to the development of a shear band. Shear-band formation in plane-strain tests has been discussed in detail in Wanatowski and Chu (2006).

The b value versus ϵ_1 curves are shown in Fig. 8d. The b value in each test increased quickly to a peak value and remained till the end of the test if strain softening did not occur. If strain softening did occur, the b value tended to decrease slightly then. The effective stress paths obtained from the above six tests are presented in Fig. 8e. The CSL_{ps} , as determined by drained plane-strain tests on very loose sand (see Fig. 7e), is also plotted in Fig. 8e. The effective stress paths from all tests ended at the CSL . The end points were at the steady state, or the critical state, because the stresses (Figs. 8a and 8b) and pore pressures (Fig. 8c) at the end points were constant, except for K0U09ps. This suggests that the same CSL was approached in both the drained and the undrained plane-strain tests.

As with the triaxial tests, a line can be drawn through the peak point of the undrained effective stress path, as shown in Fig. 8e. It can be seen that the smaller the e_c is, the steeper the slope of the line η_{IL} will be. This behaviour was consistent with that observed in triaxial tests conducted on very loose sand under undrained conditions.

Discussion

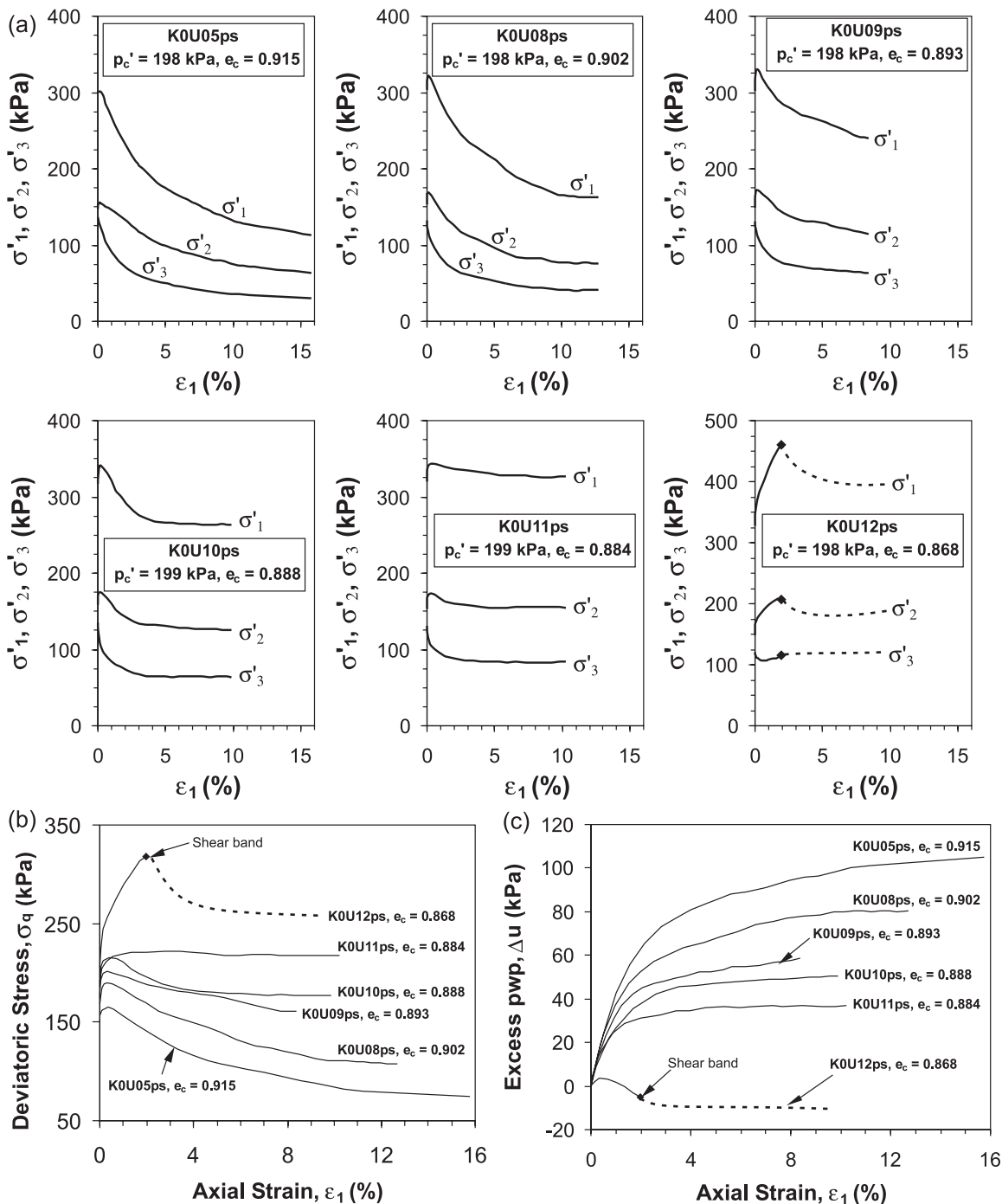
Variation in b value

The variation in b value with axial strain (ϵ_1) is compared in the drained and undrained plane-strain tests in Fig. 9. As the data have been presented in Figs. 7d and 8d, only the upper and lower bounds for each type of test are plotted in Fig. 9. The b values at the peak and at the end of the tests are also given in Table 2. Figures 7d and 8d show that the b value in each test increased to a constant peak value and remained at this value till the end of the test. However, when strain softening occurred, the b value decreased slightly. The b -value bound for the undrained tests was larger than that for the drained tests because strain softening occurred in the undrained tests. The peak b value for the undrained tests was generally larger than that for the drained tests, as shown clearly in Fig. 9. Therefore, the b value corresponding to plane-strain conditions was not unique.

Critical-state line

The CSL_{ps} obtained from CK_0 D plane-strain tests and the CSL_{tc} obtained from CID triaxial tests are compared in Fig. 10 for the Changi sand. The steady state, or critical-state, points obtained from undrained plane-strain and triaxial tests on very loose sand are also plotted in Fig. 10. All the data points in Fig. 10b, except for two triaxial test points where the effective mean stresses (p') are only 3 and 4 kPa, fall along the same CSL for both the drained and the undrained tests. Those two data points suggest that the CSL_{tc} is curved at low stress, as shown in Fig. 10b. However, more data points are needed to verify this. Figure 10a shows that the CSL obtained under plane-strain conditions was different from that obtained under axisymmetric conditions. This is not surprising, as the dependence of the critical stress ratio (M) on the intermediate principal stress, either in terms of Lode's angle (θ) or b value, was pointed out by Jefferies and Shuttle (2002). The difference in CSL for triaxial and plane-strain conditions was also observed by Finn et al. (1996). Jefferies and Shuttle (2002) used the Mohr–Coulomb and Matsuoka and Nakai (1974) failure criteria, as shown in Fig. 11a, to establish the relationships between M and θ . These relationships also established the relationship between the critical stress ratio (M) and the b value (Fig. 11b). The dotted line in Fig. 11a or Fig. 11b indicates the average M for the two criteria. The plane-strain data are also plotted in Figs. 11a and 11b. The data agree with the curves reasonably well. However, although the experimental data follow the trend predicted by Jefferies and Shuttle (2002), the data do not indicate a unique relationship between M and the b value, as shown in Fig. 11b. For example, the M value ranged from 1.1 to 1.17 for a b value of 0.27. Similar observations were made when Cornforth's (1964) data were compared with the relationship shown in Fig. 11a (Jefferies and Shuttle 2002). This implies that the M and b value (or θ) relationship may not be unique. What is less well known or much less discussed in literature is that the CSL s on the e - $\log p'$ plane, obtained under axisymmetric and plane-strain conditions, are different, as shown in Fig. 10b. Different CSL s on the e - $\log p'$ plane mean different values for the state parameter (ψ). Figure 10b shows that the two CSL s are parallel, with CSL_{tc} being below CSL_{ps} . Thus, the difference

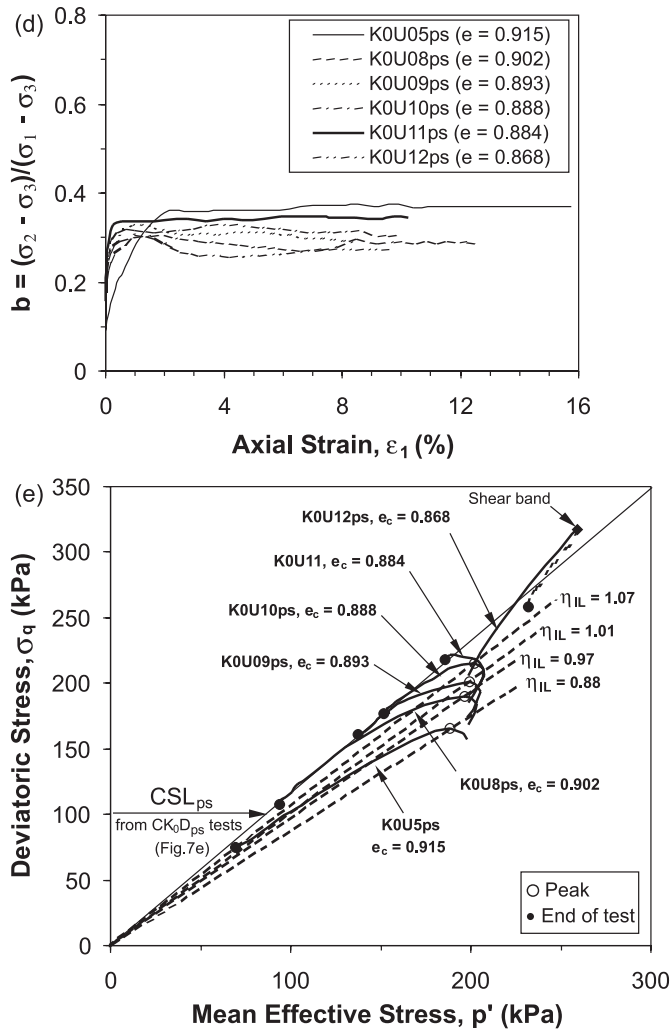
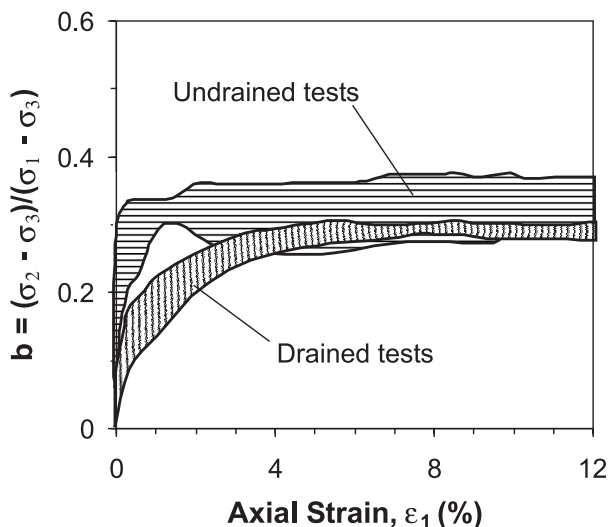
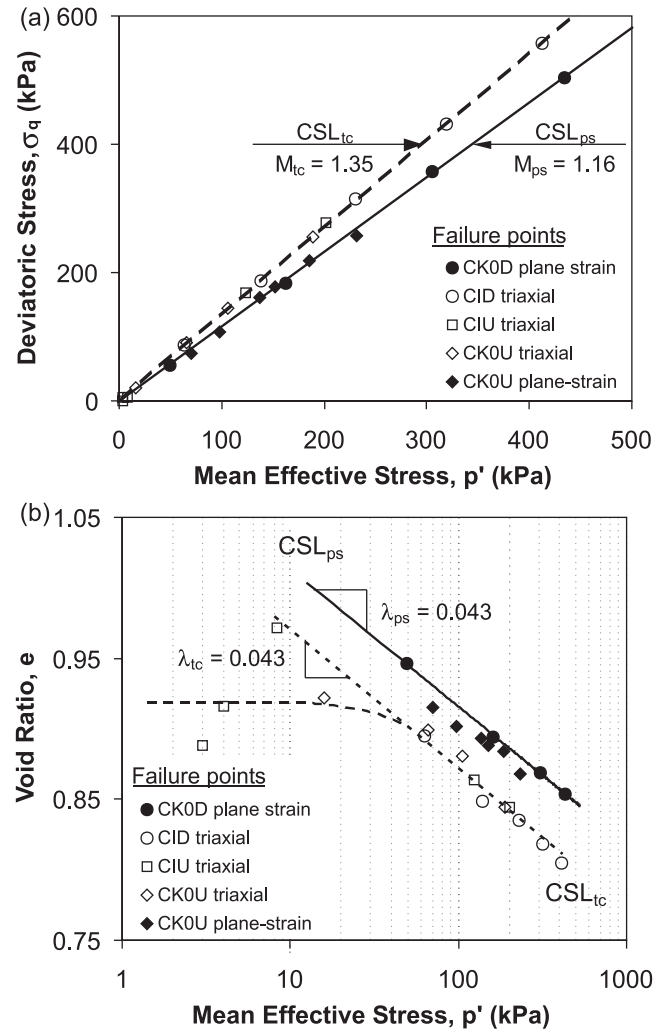
Fig. 8. Results of consolidated undrained plane-strain tests conducted on very loose sand under conditions of no lateral strains: (a) effective principal stresses; (b) stress–strain curves; (c) Δu – ε_1 curves; (d) b – ε_1 curves; (e) effective stress paths. CK₀D_{ps}, consolidated drained plane-strain tests under conditions of no lateral strains; CSL_{ps}, critical-state line under plane-strain conditions; e_c , void ratio at end of consolidation; p'_c , mean effective stress at end of consolidation; pwp, pore-water pressure; η_{IL} , effective stress ratio at peak deviatoric stress.



in the ψ values obtained from plane-strain and axisymmetric conditions can be represented by a constant value, which is about 0.05 for the data shown in Fig. 10b. This implies that the ψ value defined with respect to CSL_{tc} will be larger. Therefore, the ψ estimated by triaxial tests was on the conservative side as far as liquefaction potential is concerned.

Instability line

The instability line—that is, the line drawn through the peak point of the undrained effective stress path (see Fig. 5c)—was used to define the conditions of instability by Lade (1992, 1993). Here, instability refers to a behaviour in which large plastic strains are generated rapidly because of

Fig. 8 (concluded).**Fig. 9.** Variation of b value in plane-strain tests.**Fig. 10.** Critical-state lines for plane-strain and triaxial conditions: (a) q - p' plane; (b) e - $\log p'$ plane. CSL_{ps}, critical-state line under plane-strain conditions; CSL_{tc}, critical-state line under triaxial compression; M_{ps} , effective stress ratio at critical state for plane-strain conditions; M_{tc} , effective stress ratio at critical state for triaxial compression; λ_{ps} , slope of critical-state line under plane-strain conditions; λ_{tc} , slope of critical-state line under triaxial compression.

the inability of a soil element to sustain a given load or stress. The zone bounded by the instability line and the CSL is called the zone of instability, as loose sand will become

unstable when the stress state falls into this zone (Lade 1992, 1993). Different interpretations using the collapse-surface approach have also been given to the instability behaviour by other researchers (Sladen et al. 1985a; Sasitharan et al. 1993). Been and Jefferies (2004) argued that the collapse or instability framework has no significance in terms of effective stress models for soil liquefaction. Nevertheless, there appears to be some physical meaning behind the instability line. It can specify a yielding point where large plastic strain can develop, as explained by Lade (1993) and Chu et al. (2003). It has also been shown that the instability line is not unique but varies with the void ratio and stress level (Imam et al. 2002; Chu et al. 2003).

The instability line concept can be extended into plane-strain conditions by using the plane-strain testing data. The effective stress paths of the three CK₀U plane-strain tests

Fig. 11. Variation in critical stress ratio: (a) Lode's angle; (b) b value.

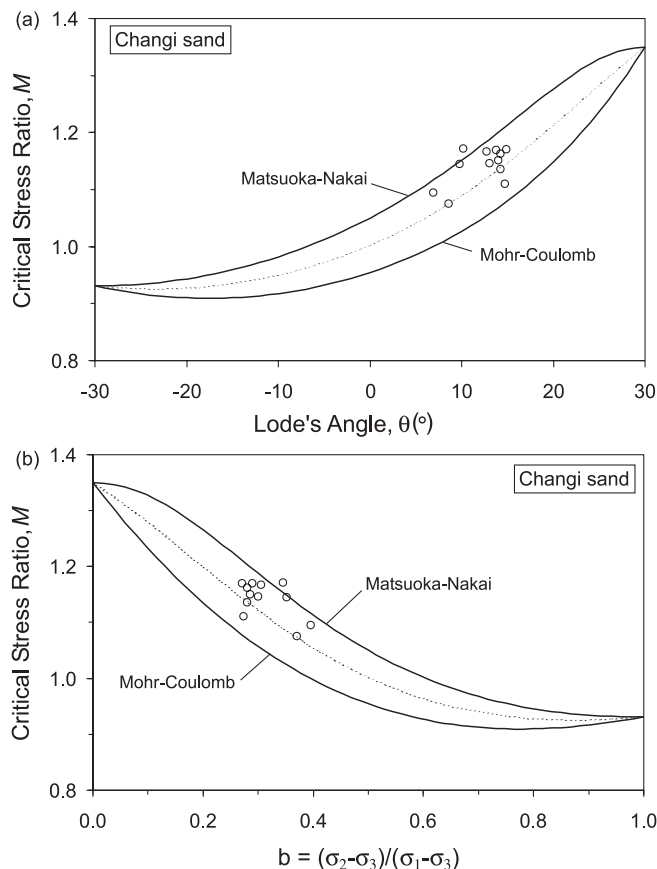
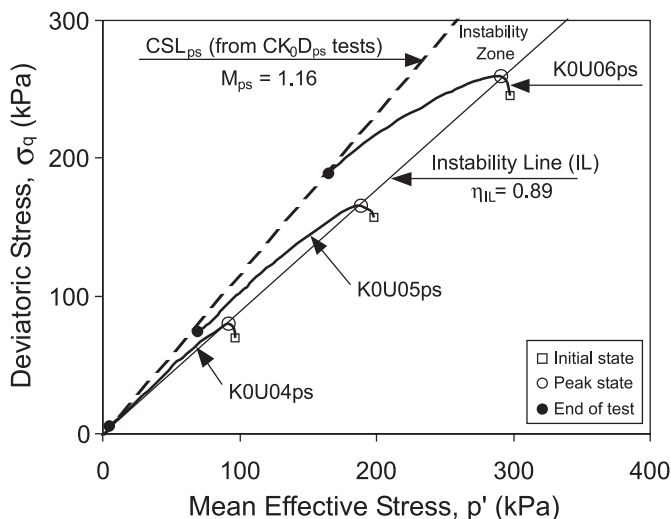


Fig. 12. Definition of instability line based on results of consolidated undrained plane-strain tests conducted on very loose sand under conditions of no lateral strains. CK_0D_{ps} , consolidated drained plane-strain tests under conditions of no lateral strains; CSL_{ps} , critical-state line under plane-strain conditions; M_{ps} , effective stress ratio at critical state for plane-strain conditions; η_{IL} , effective stress ratio at peak deviatoric stress.



(CK0U04ps, CK0U05ps, and CK0U06ps), conducted on very loose sand, are presented in Fig. 12. An instability line was determined by connecting the peak points of the undrained stress paths obtained from the CK_0U plane-strain tests. The zone of instability is defined as the zone bounded by the instability line and the CSL. As shown earlier in Fig. 8e, the effective stress ratio at the peak (η_{IL}) in plane strain was not unique but varied with the void ratio (e) of sand. This is in agreement with the results of the triaxial tests presented in Figs. 5c and 6c.

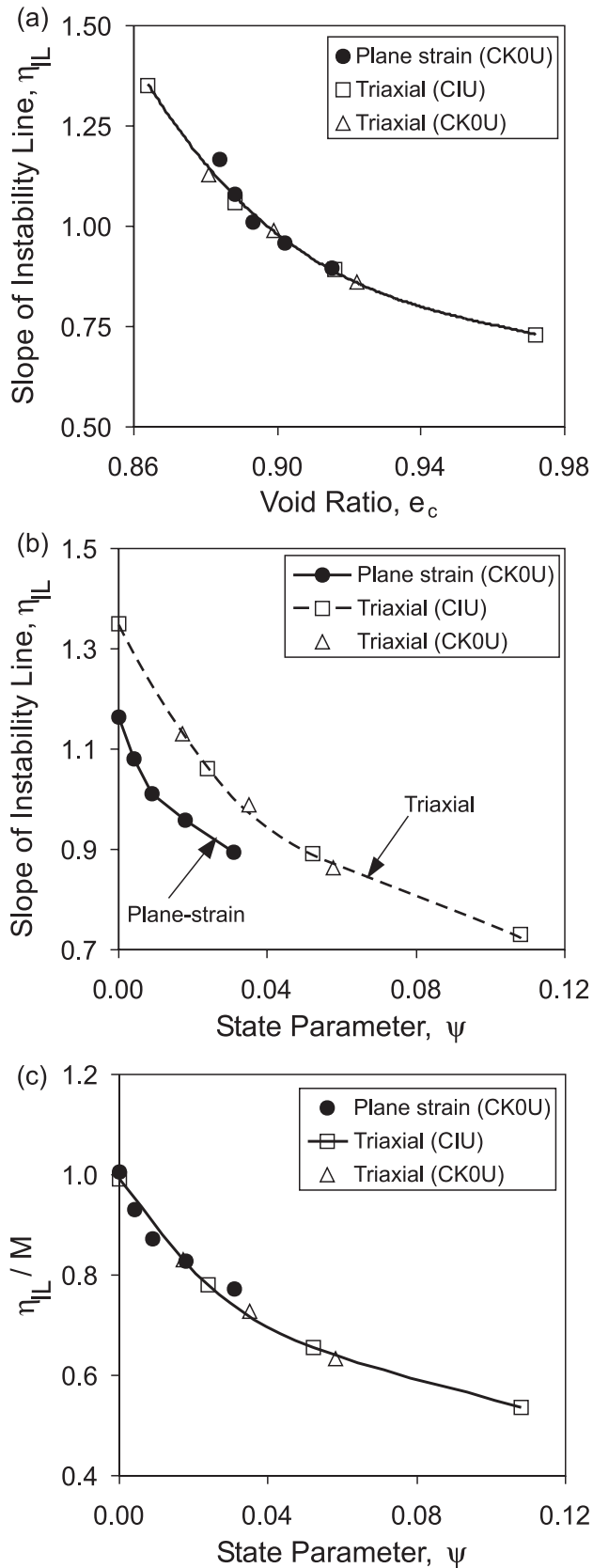
The relationship between the stress ratio of the instability line (η_{IL}) and e_c is plotted in Fig. 13a from the data obtained from both the triaxial and plane-strain tests. The relationship shown in Fig. 13a does not reflect the influence of stress level. A single relationship between η_{IL} and e_c was obtained in Fig. 13a because the mean effective stresses (p') used in Fig. 13a were within a narrow range. Figure 13b, in which η_{IL} is plotted against ψ , takes both e_c and p'_c into consideration. Here, the ψ value was measured with respect to the CSL established under axisymmetric and plane-strain conditions. Two curves were obtained: one for triaxial conditions and another for plane-strain conditions. This implies that the η_{IL} and ψ relationship was affected by the intermediate principal stress. If we plot the normalized stress ratio at peak (η_{IL}/M) against ψ (Fig. 13c), a unique relationship is obtained. Figure 13c might suggest that η_{IL}/M is a more fundamental parameter than η_{IL} . The fundamental meaning of the relationship may be related to the model proposed by Jefferies and Shuttle (2002). The practical meaning of Fig. 13c is that the findings established on static liquefaction of sand in triaxial tests can be extended to plane-strain conditions, provided the data can be interpreted in the way suggested in this paper. Therefore, in general, triaxial tests are still useful in predicting the instability of sandy slopes, even though slopes and embankments are actually plane-strain problems.

Comparison of undrained behaviour in triaxial and plane-strain tests

The results of two CK_0U tests conducted on very loose specimens under undrained conditions are presented in Fig. 14. Test K0U08ps was conducted under plane-strain conditions, whereas test K0U01 was conducted under axisymmetric conditions. The void ratios after consolidation (e_c) were 0.902 and 0.899, respectively. The q - ϵ_1 and Δu - ϵ_1 curves for the two tests are plotted in Fig. 14a. The trends of the two stress-strain curves are similar. The triaxial test yielded a slightly lower value for σ_q at the peak, but a higher Δu than in the plane-strain test. Nevertheless, the stress ratio at the peak is about the same ($\eta_{IL} = 0.89$). Figure 14b shows that the triaxial and plane-strain tests approached their own CSL at the end of the test. Similar observations were made by Finno et al. (1996). The differences in undrained behaviour in triaxial and plane-strain tests can be explained by the influence of intermediate principal stress, as discussed in Fig. 11.

Conclusions

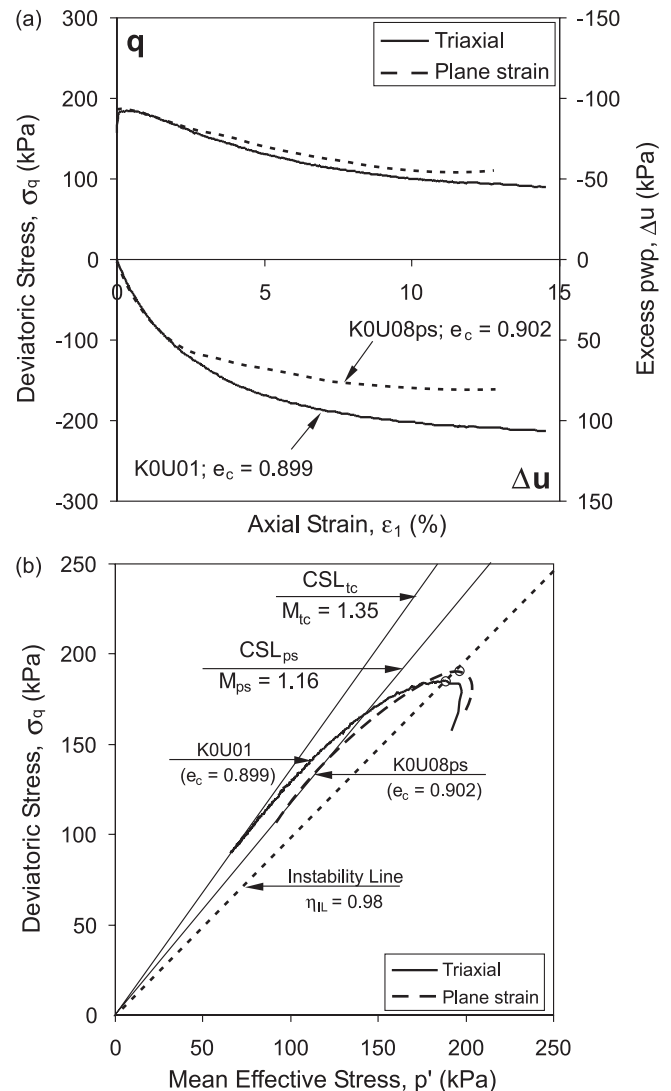
A plane-strain apparatus was used to conduct CK_0U tests to study the static liquefaction of sand under plane-strain



conditions. The undrained behaviour of sand under plane-strain and axisymmetric conditions was compared. The following conclusions can be drawn:

Fig. 13. Comparison of peak states in plane-strain and triaxial tests for $p'_c = 150\text{--}200$ kPa: (a) $\eta_{IL}\text{--}e_c$ plot; (b) $\eta_{IL}\text{--}\psi$ plot; (c) $(\eta_{IL}/M)\text{--}\psi$ plot. e_c , void ratio at end of consolidation; p'_c , mean effective stress at end of consolidation; η_{IL} , effective stress ratio at peak deviatoric stress; η_{IL}/M , normalized stress ratio at peak deviatoric stress; ψ , state parameter.

Fig. 14. Comparison of plane-strain and triaxial tests conducted on very loose sand: (a) $q\text{--}\epsilon_1$ and $\Delta u\text{--}\epsilon_1$ curves; (b) effective stress paths. CSL_{ps}, critical-state line under plane-strain conditions; CSL_{tri}, critical-state line under triaxial conditions; e_c , void ratio at end of consolidation; M_{ps} , effective stress ratio at critical state for plane-strain conditions; M_{tri} , effective stress ratio at critical state for triaxial conditions; pwp, pore-water pressure; η_{IL} , effective stress ratio at peak deviatoric stress.



- The b value corresponding to a plane-strain condition was not unique. It varied during a plane-strain test, and the peak b value varied from test to test. The peak b value obtained from undrained plane-strain tests was in general larger than that obtained from drained plane-strain tests.
- The undrained behaviour of very loose sand under plane-strain conditions was similar to that under axisymmetric

conditions. A slightly higher peak deviatoric stress was obtained in the plane-strain tests, whereas a higher excess pore water pressure was observed in the triaxial tests. As a result, triaxial tests provide a conservative estimate in predicting the liquefaction potential of soils.

- On both the q - p' and e - p' planes, the CSL obtained from the plane-strain tests was different from that obtained from the triaxial tests. The slope of the CSL (M) was dependent on the intermediate principal stress, and the state parameter (ψ) for plane strain was about 0.05 less than that for axisymmetric conditions for the sand tested.
- The instability line was defined under plane-strain conditions by using the method suggested by Lade (1992) for axisymmetric conditions. Like the observations made under axisymmetric conditions, the instability line was not unique, but varied with the void ratio of the sand: the smaller the e_c , the steeper the slope of the instability line. A unique relationship between the slope of the instability line η_{IL} and e_c or between η_{IL}/M and ψ can be established for both plane-strain and triaxial tests. Therefore, the findings on static liquefaction of sand established in triaxial tests can be extended to the plane-strain conditions provided the data are interpreted by using the η_{IL}/M versus ψ relationship.

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References

- Alarcon-Guzman, A., Leonards, G.A., and Chameau, J.L. 1988. Undrained monotonic and cyclic strength of sands. *Journal of the Geotechnical Engineering Division, ASCE*, **114**(10): 1089–1109.
- Andresen, A., and Bjerrum, L. 1976. Slides in subaqueous slopes in loose sand and silt. In *Proceedings of the International Research Conference on Marine Geotechnique*. University of Illinois Press, Chicago. pp. 221–237.
- Been, K., and Jefferies, M.G. 1985. A state parameter for sands. *Géotechnique*, **35**(2): 99–112.
- Been, K., and Jefferies, M.G. 2004. Stress–dilatancy in very loose sand. *Canadian Geotechnical Journal*, **41**(5): 972–989.
- Been, K., Jefferies, M.G., and Hachey, J.E. 1991. The critical state of sands. *Géotechnique*, **41**(3): 365–381.
- Casagrande, A. 1936. Characteristics of cohesionless soils affecting the stability of slopes and earth fills. *Journal of the Boston Society of Civil Engineers, ASCE*, **23**(1): 13–32. Reprinted 1940. In *Contributions to soil mechanics: 1925–1940*. Boston Society of Civil Engineers, Boston, Mass. pp. 257–276.
- Casagrande, A. 1965. Role of calculated risk in earthwork and foundation engineering. *Journal of the Soil Mechanics and Foundations Division, ASCE*, **91**(SM4): 1–40.
- Castro, G. 1969. Liquefaction of sands. Ph.D. thesis, Harvard University, Cambridge, Mass. Harvard Soil Mechanics Series, No. 81.
- Chu, J. 1995. An experimental examination of the critical state and other similar concepts for granular soils. *Canadian Geotechnical Journal*, **32**: 1065–1075.
- Chu, J., and Gan, C.L. 2004. Effect of void ratio on K_0 of loose sand. *Géotechnique*, **54**(4): 285–288.
- Chu, J., and Leong, W.K. 2002. Effect of fines on instability behaviour of loose sand. *Géotechnique*, **52**(10): 751–755.
- Chu, J., and Lo, S.-C.R. 1993. On the measurement of critical state parameters of dense granular soils. *Geotechnical Testing Journal, ASTM*, **16**(1): 27–35.
- Chu, J., Lo, S.-C.R., and Lee, I.K. 1996. Strain softening and shear band formation of sand in multi-axial testing. *Géotechnique*, **46**(1): 63–82.
- Chu, J., Leroueil, S., and Leong, W.K. 2003. Unstable behaviour of sand and its implication for slope stability. *Canadian Geotechnical Journal*, **40**: 873–885.
- Cornforth, D.H. 1964. Some experiments on the influence of strain conditions on the strength of sand. *Géotechnique*, **14**(2): 143–167.
- Desrues, J., and Viggiani, G. 2004. Strain localization in sand: an overview of the experimental results obtained in Grenoble using stereophotogrammetry. *International Journal for Numerical and Analytical Methods in Geomechanics*, **28**: 279–321.
- Finno, R.J., and Rechenmacher, A.L. 2003. Effects of consolidation history on critical state of sand. *Journal of Geotechnical and Geoenvironmental Engineering, ASCE*, **129**(4): 350–360.
- Finno, R.J., Harris, W.W., and Viggiani, G. 1996. Strain localization and undrained steady state of sand. *Journal of Geotechnical Engineering, ASCE*, **122**(6): 462–473.
- Finno, R.J., Harris, W.W., Mooney, M.A., and Viggiani, G. 1997. Shear bands in plane strain compression of loose sand. *Géotechnique*, **47**(1): 149–165.
- Fourie, A.B., Blight, G.E., and Papageorgiou, G. 2001. Static liquefaction as a possible explanation for the Merriespruit tailings dam failure. *Canadian Geotechnical Journal*, **38**: 707–719.
- Green, P.A., and Ferguson, P.A.S. 1971. On liquefaction phenomena, by Professor A. Casagrande: report of lecture. *Géotechnique*, **21**(3): 197–202.
- Han, C., and Vardoulakis, I.G. 1991. Plane-strain compression experiments on water-saturated fine-grained sand. *Géotechnique*, **41**(1): 49–78.
- Hazen, A. 1918. A study of the slip in the Calaveras Dam. *Engineering News-Record*, **81**(26): 1158–1164.
- Hird, C.C., and Hassona, F.A.K. 1990. Some factors affecting the liquefaction and flow of saturated sands in laboratory tests. *Engineering Geology*, **28**: 149–170.
- HMSO. 1967. Report of the Tribunal appointed to inquire into the disaster at Aberfan on October 21st, 1966. Her Majesty's Stationery Office, Wales, UK.
- Imam, S.M.R., Morgenstern, N.R., Robertson, P.K., and Chan, D.H. 2002. Yielding and flow liquefaction of loose sand. *Soils and Foundations*, **42**(3): 19–31.
- Ishihara, K. 1993. Liquefaction and flow failure during earthquakes. *Géotechnique*, **43**(3): 351–415.
- Jefferies, M.G., and Shuttle, D.A. 2002. Dilatancy in general Cambridge-type models. *Géotechnique*, **52**(9): 625–638.
- Kleiner, D.E. 1976. Design and construction of an embankment dam to impound gypsum wastes. In *Proceedings of the 12th International Congress on Large Dams, Mexico City, 29 March – 1 April 1976*. International Commission on Large Dams, Paris. Vol. 1, pp. 235–249.
- Lade, P.V. 1992. Static instability and liquefaction of loose fine sandy slopes. *Journal of Geotechnical Engineering, ASCE*, **118**(1): 51–71.
- Lade, P.V. 1993. Initiation of static instability in the submarine Nerlerk berm. *Canadian Geotechnical Journal*, **30**: 895–904.

- Leong, W.K., and Chu, J. 2002. Effect of undrained creep on instability behaviour of loose sand. *Canadian Geotechnical Journal*, **39**: 1399–1405.
- Leong, W.K., Chu, J., and The, C.I. 2000. Liquefaction and instability of a granular fill material. *Geotechnical Testing Journal*, ASTM, **23**(2): 178–192.
- Lo, S.-C.R., and Chu, J. 1991. The measurement of K_0 by triaxial strain path testing. *Soils and Foundations*, **31**(2): 181–187.
- Lo, S.-C.R., Chu, J., and Lee, I.K. 1989. A technique for reducing membrane penetration and bedding errors. *Geotechnical Testing Journal*, ASTM, **12**(4): 311–316.
- Matsuoka, H., and Nakai, T. 1974. Stress-deformation and strength characteristics of soil under different principal stresses. In *Proceedings of the Japanese Society of Civil Engineers* 1974. Vol. 232, pp. 59–70.
- Mokni, M., and Desrues, J. 1999. Strain localization measurements in undrained plane-strain biaxial tests on Hostun RF sand. *Mechanics of Cohesive-frictional Materials*, **4**(4): 419–441.
- Negussey, D., and Islam, M.S. 1994. Uniqueness of steady state and liquefaction potential. *Canadian Geotechnical Journal*, **31**: 132–139.
- Olson, S.M., and Stark, T.D. 2003. Yield strength ratio and liquefaction analysis of slopes and embankments. *Journal of Geotechnical and Geoenvironmental Engineering*, ASCE, **129**(8): 727–737.
- Poulos, S.J. 1981. The steady state of deformation. *Journal of the Geotechnical Engineering Division*, ASCE, **107**(GT5): 553–562.
- Rowe, P.W., and Barden, L. 1964. Importance of free ends in triaxial testing. *Journal of the Soil Mechanics and Foundations Division*, ASCE, **90**(SM1): 1–15.
- Sasitharan, S., Robertson, P.K., Sego, D.C., and Morgenstern, N.R. 1993. Collapse behaviour of sand. *Canadian Geotechnical Journal*, **30**: 569–577.
- Schofield, A.N., and Wroth, C.P. 1968. *Critical state soil mechanics*, McGraw-Hill, London.
- Skempton, A.W. 1986. Standard penetration test procedures and the effects in sands of overburden pressure, relative density, particle size, ageing, and overconsolidation. *Géotechnique*, **36**: 425–447.
- Sladen, J.A., D'Hollander, R.D., and Krahn, J. 1985a. The liquefaction of sands, a collapse surface approach. *Canadian Geotechnical Journal*, **22**: 564–578.
- Sladen, J.A., D'Hollander, R.D., Krahn, J., and Mitchell, D.E. 1985b. Back analysis of the Nerlerk berm liquefaction slides. *Canadian Geotechnical Journal*, **22**: 579–588.
- Tatsuoka, F., Sakamoto, M., Kawamura, T., and Fukushima, S. 1986. Strength and deformation characteristics of sand in plane strain compression at extremely low pressures. *Soils and Foundations*, **26**(1): 65–84.
- Topolnicki, M., Gudehus, G., and Mazurkiewicz, B.K. 1990. Observed stress-strain behaviour of remoulded saturated clay under plane-strain conditions. *Géotechnique*, **42**(2): 155–187.
- Verdugo, R., and Ishihara, K. 1996. The steady state of sandy soils. *Soils and Foundations*, **36**(2): 81–91.
- Wanatowski, D., and Chu, J. 2006. Stress-strain behavior of a granular fill measured by a new plane-strain apparatus. *Geotechnical Testing Journal*, ASTM, **29**(2): 149–157.
- Yamamuro, J.A., and Lade, P.V. 1997. Static liquefaction of very loose sands. *Canadian Geotechnical Journal*, **34**: 905–917.
- Yang, J. 2002. Non-uniqueness of flow liquefaction line for loose sand. *Géotechnique*, **52**(10): 757–760.

List of symbols

- b parameter that measures relative magnitude of intermediate principal stress ($= (\sigma_2 - \sigma_3)/(\sigma_1 - \sigma_3)$)
- b_f b at failure
- b_p b at peak
- B Skempton's pore-pressure parameter
- e void ratio
- e_c void ratio at end of consolidation
- e_{cs} void ratio at critical state
- e_f void ratio at failure
- e_p void ratio at peak
- K_0 coefficient of lateral earth pressure at rest
- M η at critical state (M_{ps} in plane-strain tests; M_{tc} in triaxial tests)
- p' mean effective stress in plane-strain tests ($= (\sigma'_1 + \sigma'_2 + \sigma'_3)/3$) and in triaxial tests ($= (\sigma'_1 + 2\sigma'_3)/3$)
- p'_c mean effective stress at end of consolidation
- p'_f mean effective stress at failure
- p'_p mean effective stress at peak
- q deviatoric stress
- q_c deviatoric stress at end of consolidation
- q_f deviatoric stress at failure
- q_p deviatoric stress at peak
- u pore-water pressure
- Δu excess pore-water pressure
- ϵ_1 axial strain
- ϵ_v volumetric strain (compression positive)
- η effective stress ratio ($= q/p'$)
- η_{ll} η at peak deviatoric stress
- θ Lode's angle ($= -\tan^{-1}[(2b - 1)/(3^{1/2})]$)
- λ slope of the CSL on $e - \log p'$ plane
- $\sigma'_1, \sigma'_2, \sigma'_3$ major, intermediate, and minor principal effective stresses
- σ_q deviatoric stress in plane-strain tests ($= [(\sigma'_1 - \sigma'_2)^2 + (\sigma'_2 - \sigma'_3)^2 + (\sigma'_3 - \sigma'_1)^2]^{1/2}/(2^{1/2})$) and in triaxial tests ($= q = \sigma'_1 - \sigma'_3$)
- ϕ effective friction angle of soil
- ϕ_{cs} ϕ at critical state ($(\phi_{cs})_{ps}$ in plane-strain test; $(\phi_{cs})_{tc}$ in triaxial test)
- ψ state parameter ($= e - e_{cs}$)